SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	M.Deepika	**Reg. No.:**	192372296
Section / Batch:	2023	**Date:**	15/07/2026

Code of Conduct

I ________________M.Deepika______192372296____________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: __deepika.m_____________________

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation. • Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
	GRAND TOTAL:	**100 Marks**	

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks) (OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, wellstructured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

Introduction to Linked Lists for Flexible Playlist Structures

Understanding Linked List Fundamentals

Linked lists offer a robust solution for dynamic data management, providing flexibility in handling additions, removals, and reordering operations.

Enhancing User Experience with Flexibility

Leveraging linked lists directly contributes to an enhanced user experience by enabling seamless and responsive playlist modifications.

Optimizing Dynamic Song Management

This section explores how linked lists are optimally suited for dynamic song management within music playlists, ensuring efficient operations.

Efficient Song Insertion and Deletion in Playlists

Linked List Insertion Mechanics

Inserting a song into a linked list involves updating pointers, a constant-time operation that maintains efficiency regardless of playlist size.

Streamlining Playlist Additions and Removals

The inherent structure of linked lists streamlines the process of adding and removing songs, making these operations highly efficient.

Seamless Playlist Modification for Users

Efficient insertion and deletion directly translate to a seamless playlist modification experience, minimizing latency for users.

Reordering Songs Seamlessly within a Linked List

Linked List Reordering Techniques

Reordering in a linked list typically involves adjusting pointers to change the sequence of nodes, an operation that is generally efficient.

Reordering Songs in Playlists

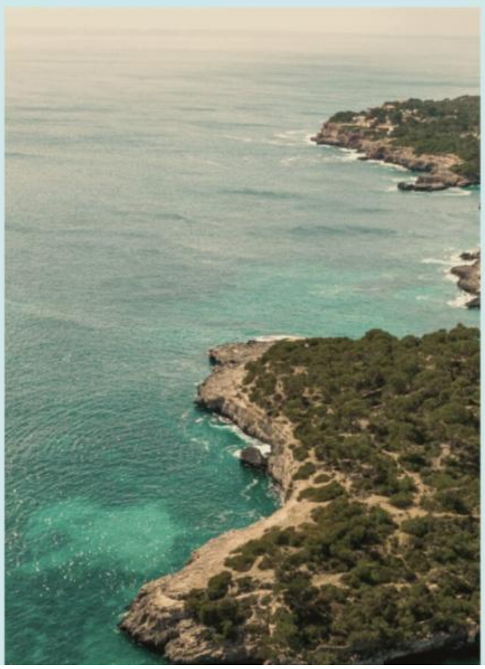
The dynamic reordering of songs is a key feature for music playlists, allowing users to customize their listening experience effectively.

Optimizing Song Sequence Adjustments

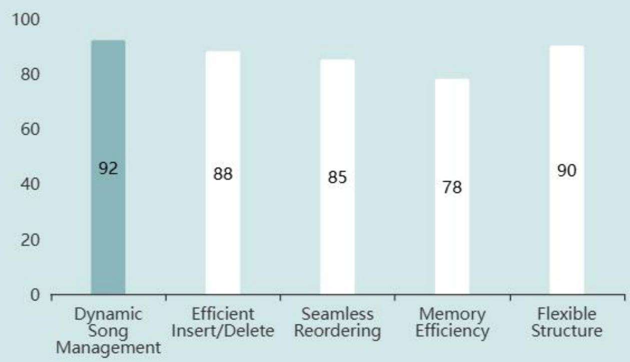
Linked lists provide an optimized approach for adjusting song sequences, ensuring that reordering operations are performed with minimal computational overhead.

Improving User Control and Customization

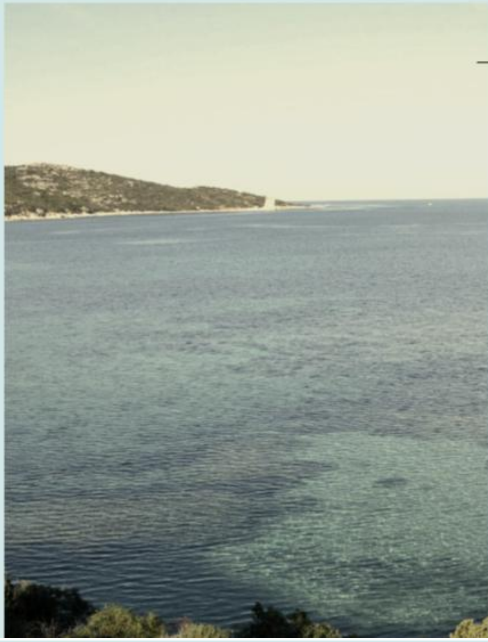
Seamless song reordering empowers users with greater control over their playlists, significantly enhancing their overall application experience.



Linked List Advantages for Dynamic Music Playlists



Key Benefits of Linked Lists in Playlist Management(Effectiveness Score (%))



2 Performance Analysis and Justification of Linked List Implementation for Playlists

Comparative Efficiency of Data Structures for Playlists

Linked List for Playlist Management

A doubly linked list is highly recommended for dynamic playlist management due to its inherent flexibility and efficient performance for common playlist operations.

Optimizing User Interaction

The choice of data structure directly impacts the responsiveness and fluidity of user interactions within the music streaming application, making efficiency paramount.

Real-world Application and Benefits of Linked Lists in Music Streaming

Practical Playlist Implementation

In real-world music streaming applications, the dynamic nature of user-created playlists necessitates a data structure that can adapt efficiently to frequent

Enhanced User Experience

The flexibility and efficiency of linked lists directly contribute to an enhanced user experience, allowing for fluid playlist manipulation without noticeable lag.

Advantages in Music Streaming

Linked lists provide significant advantages in music streaming by facilitating quick song reordering and insertion/deletion, crucial for dynamic playlist management

Justification for Linked Lists

This section provides a comprehensive justification for the selection of linked lists, detailing their practical benefits and operational efficiency in a music streaming environment.

Comparative Efficiency of Data Structures for Playlists

Efficiency of Data Structures in Playlist Management(Efficiency Score (%))

THANK YOU

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